Nucleation, Freezing Range, and Hot Cracking in Aluminum Alloys Solidified at Laser Melting Rates

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NOFO Number: DE-FOA-0003665

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Research area	Basic Energy Sciences, Materials Sciences and Engineering Division, Synthesis and Processing Science (NOFO Part 1, Section III)
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1. Scientific questions

Most high-strength aluminum alloys, such as 7075, are unsuitable for laser powder bed fusion (LPBF): printing gives large columnar grains and periodic cracks [1]. Martin et al. removed both by assembling zirconium-based nanoparticles onto the powder, which gave crack-free, equiaxed, fine-grained material [1]. That result leaves the mechanism open. We ask three questions. (1) Why does a given addition of zirconium or scandium give equiaxed grains under some laser conditions and columnar grains under others? (2) Why does cracking follow the shape of the solidification curve near its end [2], and how do grain shape and the loss of magnesium and zinc by evaporation [3] change that relation? (3) How far do equilibrium and Scheil calculations of phase diagrams (CALPHAD) depart from what actually forms when aluminum solidifies at LPBF speeds?

The laser is a moving, localized heat source. The thermal gradient G and solidification speed R it imposes at the back of the melt pool direct how grains grow, and both change with laser power and scan speed. A single build plate can therefore hold dozens of single tracks, each solidified under different conditions that we can estimate. These questions concern nucleation and phase transitions far from equilibrium, and the answers apply to welding and other rapid solidification processes as well as to printing.

2. Hypotheses and tests

H1, nucleation. Equiaxed grains form only if primary Al_3Zr or $Al_3(Sc,Zr)$ particles precipitate in the cooling liquid before the aluminum grows past them. At higher scan speed the liquid spends less time in that window, so fewer particles act as nucleants. Prediction: the smallest Zr + Sc content that gives equiaxed grains rises with scan speed, and the boundary follows Hunt's criterion for the columnar-to-equiaxed transition [4], with the nucleant density computed from the CALPHAD primary-particle fraction. H1 is rejected if that critical content does not change with scan speed beyond the scatter of replicate tracks.

H2, cracking. Crack density at LPBF speeds is set by Kou's index, the steepness of temperature against the square root of fraction solid near the end of solidification [2], when the index is computed for the composition that actually solidifies, after evaporation. Equiaxed grains raise the index at which cracks first appear. Prediction: crack density against the index falls on two separate curves, one for columnar and one for equiaxed tracks, and the measured composition ranks cracking better than the nominal one. H2 is rejected if neither grain shape nor the composition correction improves the rank correlation between index and crack density.

H3, departure from CALPHAD. At LPBF solidification speeds the growing solid traps solute and some intermetallic phases do not form. Prediction: measured freezing behavior and phase fractions lie between equilibrium and Scheil predictions at low scan speed and move outside the Scheil bound as speed rises. H3 is rejected if measured phase fractions and segregation stay within the Scheil bound at every speed tested.

3. Approach

Alloys and powder. We will study an Al-Zn-Mg-Cu base near 7075 with zirconium and scandium additions over a range of contents. Powder for each composition is made in our laboratory by an ultrasonic atomizer (AMAZEMET rePOWDER) with an induction-melting module, at four compositions per week. Making our own powder lets us change composition in small steps each week.

Controlled solidification. LPBF builds are bought on a purchase order from a University of Utah user facility at $800 per build. Each plate holds single tracks over a grid of laser power and scan speed, plus small cubes at selected settings, with replicates to measure scatter. We will estimate G and R for each track from its melt-pool cross-section and scan speed, calibrated with a thermal model.

Measurements. At the BYU Microscopy Center, scanning electron microscopy (SEM) gives melt-pool shape and crack density, electron backscatter diffraction (EBSD) gives grain shape and the equiaxed fraction,

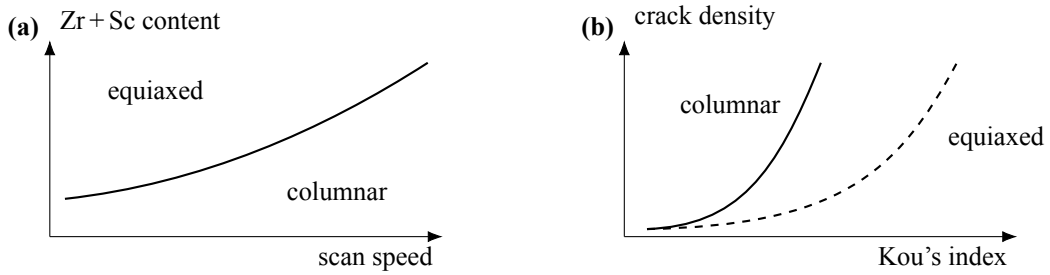


Figure 1. Sketched predictions. (a) H1: the nucleant content needed for equiaxed grains rises with scan speed. (b) H2: equiaxed tracks crack at a higher index than columnar tracks of the same alloy.

and energy-dispersive X-ray spectroscopy (EDS) gives magnesium and zinc loss and solute segregation across cells. Tensile and hardness tests on selected compositions connect grain shape to strength and ductility.

Calculations. Equilibrium and Scheil CALPHAD calculations run on BYU high-performance computing for every composition, on both the nominal and the measured composition. They supply the primary-particle fraction and formation temperature (H1), Kou's index (H2), and the reference phase fractions that H3 tests.

Choosing experiments. The composition and laser space is too large to measure on a grid, and the measurements differ in cost from minutes (calculations) to days (tensile tests). Statistical models linked in the same order as the steps [5] let us choose the next compositions and laser settings where H1 to H3 and their alternatives predict the most different outcomes per unit cost. Each composition then stops at the cheapest measurement that answers the question [6]: most stop at tracks, and few reach tensile tests. We are extending this method to several outputs at once and to results that arrive at different times. In this project its role is to place experiments near the transitions in H1 and H2.

Alternatives. Nucleant particles may survive from the powder instead of forming in the melt pool. A second laser pass over the same track separates the two: particles carried over from the powder are lost on remelting, while particles that form in the cooling liquid form again. If cracking does not follow Kou's index at all, we will compare crack density with the CALPHAD freezing range and with the fraction of liquid left at the end of solidification, which are older measures of cracking tendency.

Plan. Year 1 tests H3 and H1 on tracks for zirconium-only alloys over a range of scan speeds, including remelting tests. Year 2 adds scandium and zirconium plus scandium, builds cubes, and tests H2. Year 3 tests the combined rule on new compositions chosen to challenge it, adds tensile tests, and releases the data.

4. Expected results and team

The project will produce measured maps of grain shape and cracking across composition, G, and R for Al-Zn-Mg-Cu alloys with Zr and Sc. It will also give a tested rule for when a nucleant addition gives equiaxed grains at high solidification speed, measured departures from equilibrium and Scheil predictions at LPBF speeds, and open data and code. Results that reject a hypothesis will be published like any other.

Since the laboratory opened at Brigham Young University (BYU) in December 2025, we have set up the CALPHAD workflow on BYU high-performance computing and completed a powder-to-part demonstration on a baseline aluminum alloy: atomized powder, LPBF builds, SEM and EDS, and tensile data. In a hand-built, noise-free simulation in which a CALPHAD crack screen ignored zirconium, choosing each next measurement by expected gain per cost matched the best result of screening first after 3.8 times less spending (90% confidence interval 2.0 to 5.7). The PI (Ph.D., Materials Science and Engineering, University of Utah, 2023) inferred grain boundary models with propagated uncertainty, developed Honegumi for setting up Bayesian optimization, and co-authored the *Chemical Reviews* survey of self-driving laboratories. The PI will supervise one Ph.D. student and two part-time undergraduate researchers over three years.

References

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